AS-P1-CAL-001 Rev 0. Maintenance Platform Hand Calculation. Abbas Saleh, Oct 17 2026. Student exercise, not for construction.

Joist J1

$LJ := 3 \cdot m$
$sJ := 1.5 \cdot m$
$qD := 0.5 \cdot kPa$
$qL := 4.8 \cdot kPa$
$mJ := 22.5 \ \frac{kg}{m}$

$E := 200000 \cdot MPa$
$Sx := 159 \cdot 10^3 \cdot mm^3$
$Ix := 12.1 \cdot 10^6 \cdot mm^4$
$d := 152 \cdot mm$
$tw := 5.8 \cdot mm$
$wsw := mJ \cdot g$
$wD := qD \cdot sJ + wsw$
$wL := qL \cdot sJ$
$wF := 1.25 \cdot wD + 1.5 \cdot wL$

$$Mf := wF \cdot \frac{LJ^2}{8}$$
$$Vf := wF \cdot \frac{LJ}{2}$$
$$\sigma := \frac{Mf}{Sx}$$
$$\tau := \frac{Vf}{(d \cdot tw)}$$

$ws := wD + wL$

$$\delta := 5 \cdot ws \cdot \frac{LJ^4}{(384 \cdot E \cdot Ix)}$$

$$\delta lim := \frac{LJ}{360}$$

$Mf = 13.515 \ kN \cdot m$
$Vf = 18.02 \ kN$
$\sigma = 85 \ MPa$
$\tau = 20.44 \ MPa$
$\delta = 3.561 \ mm$
$\delta lim = 8.333 \ mm$

Main beam B1

$LB := 6 \cdot m$

$mB := 38.8 \cdot \frac{kg}{m}$

$SxB := 549 \cdot 10^3 \cdot mm^3$

$IxB := 85.1 \cdot 10^6 \cdot mm^4$

$dB := 310 \cdot mm$

$twB := 5.8 \cdot mm$

$Peq := 10 \cdot kN$

$wswB := mB \cdot g$

$PD := wD \cdot \frac{LJ}{2}$

$PL := wL \cdot \frac{LJ}{2}$

$Pf := 1.25 \cdot PD + 1.5 \cdot PL$

$Peqf := 1.5 \cdot Peq$

$wBf := 1.25 \cdot wswB$

$RB := \frac{(3 \cdot Pf + Peqf)}{2} + wBf \cdot \frac{LB}{2}$

$MB := RB \cdot 0.5 \cdot LB - Pf \cdot 1.5 \cdot m - 0.125 \cdot wBf \cdot LB^2$

$\sigma B := \frac{MB}{SxB}$

$\tau B := \frac{RB}{(dB \cdot twB)}$

$Ps := PD + PL$

$a1 := 1.5 \cdot m$

$\delta 1 := 2 \cdot Ps \cdot a1 \cdot \frac{\langle 3 \cdot LB^2 - 4 \cdot a1^2 \rangle}{(48 \cdot E \cdot IxB)}$

$\delta 2 := (Ps + Peq) \cdot \frac{LB^3}{(48 \cdot E \cdot IxB)}$

$\delta 3 := 5 \cdot wswB \cdot \frac{LB^4}{(384 \cdot E \cdot IxB)}$

$\delta B := \delta 1 + \delta 2 + \delta 3$

$\delta Blim := \frac{LB}{360}$

$RB = 35.957\ \boldsymbol{kN}$
$MB = 78.7\ \boldsymbol{kN \cdot m}$
$\sigma B = 143.352\ \boldsymbol{MPa}$
$\tau B = 19.998\ \boldsymbol{MPa}$
$\delta B = 10.717\ \boldsymbol{mm}$
$\delta Blim = 16.667\ \boldsymbol{mm}$

Post C1

$HC := 3 \cdot \boldsymbol{m}$
$K := 1$
$AC := 2960 \cdot \boldsymbol{mm}^2$
$IC := 7.05 \cdot 10^6 \cdot \boldsymbol{mm}^4$

$mC := 23.2 \cdot \dfrac{\boldsymbol{kg}}{\boldsymbol{m}}$

$wDe := qD \cdot 0.75 \cdot \boldsymbol{m} + wsw$
$wLe := qL \cdot 0.75 \cdot \boldsymbol{m}$

$Re := \left(1.25 \cdot wDe + 1.5\ wLe\right) \cdot 0.5 \cdot LJ$
$PC := RB + Re + 1.25 \cdot mC \cdot g \cdot HC$

$Pcr := \pi^2 \cdot E \cdot \dfrac{IC}{\left(K \cdot HC\right)^2}$

$\sigma C := \dfrac{PC}{AC}$

$ratioC := \dfrac{PC}{Pcr}$

$PC = 46.027\ \boldsymbol{kN}$
$Pcr = \left(1.546 \cdot 10^3\right)\ \boldsymbol{kN}$
$\sigma C = 15.55\ \boldsymbol{MPa}$
$ratioC = 0.03$

INPUT AS-P1-INP-001 Input Sheet

GEOMETRY	Symbol	Value	Unit	Source / assumption
Main beam span (grid 1 to 2)	L_B1	6	m	Assumed scenario
Joist span (grid A to B)	L_J1	3	m	Assumed scenario
Joist spacing	s_J	1.5	m	Assumed scenario
Number of joists	n_J	5	-	At 0, 1.5, 3.0, 4.5, 6.0 m
Post height (grade to top of steel)	H_C1	3	m	Assumed scenario
Post effective length factor	K	1	-	Assumed: braced frame, pinned-pinned posts
Equipment load position on B1	x_eq	3	m	Midspan of B1

MATERIAL	Symbol	Value	Unit	Source / assumption
Yield strength	Fy	350	MPa	Assumed CSA G40.21 350W
Elastic modulus	E	200000	MPa	Standard value for structural steel
Gravity	g	9.81	m/s2	Standard

SECTION PROPERTIES (fill from CISC database)	Symbol	Value	Unit	Source / assumption
B1 W310x39: area	A	4940	mm2	beamdimensions.com, CISC W310x39
B1 W310x39: depth	d	310	mm	beamdimensions.com, CISC W310x39
B1 W310x39: web thickness	w	5.8	mm	beamdimensions.com, CISC W310x39
B1 W310x39: moment of inertia (strong axis)	Ix	85.1	x10^6 mm4	beamdimensions.com, CISC W310x39
B1 W310x39: elastic section modulus	Sx	549	x10^3 mm3	beamdimensions.com, CISC W310x39
B1 W310x39: mass	m	38.8	kg/m	beamdimensions.com, CISC W310x39
B1 self-weight (formula)	w_sw	0.380628	kN/m	Formula: mass x g / 1000
J1 W150x22: area	A	2860	mm2	beamdimensions.com, CISC W150x22
J1 W150x22: depth	d	152	mm	beamdimensions.com, CISC W150x22
J1 W150x22: web thickness	w	5.8	mm	beamdimensions.com, CISC W150x22
J1 W150x22: moment of inertia (strong axis)	Ix	12.1	x10^6 mm4	beamdimensions.com, CISC W150x22

J1 W150x22: elastic section modulus	Sx	159	x10^3 mm3	beamdimensions.com, CISC W150x22
J1 W150x22: mass	m	22.5	kg/m	beamdimensions.com, CISC W150x22
J1 self-weight (formula)	w_sw	0.220725	kN/m	Formula: mass x g / 1000
C1 HSS127x127x6.4: area	A	2960	mm2	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: moment of inertia	I	7.05	x10^6 mm4	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: radius of gyration	r	48.8	mm	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: mass	m	23.2	kg/m	CISC HSS table (search HSS127x127x6.4 on the same site)
BR1 L64x64x6.4: area	A	766	mm2	CISC angle table (bracing: model + MTO only, not designed)
BR1 L64x64x6.4: mass	m	6.01	kg/m	CISC angle table (bracing: model + MTO only, not designed)

LOADS	**Symbol**	**Value**	**Unit**	**Source / assumption**
Grating (dead)	q_D	0.5	kPa	Assumed
Floor live load	q_L	4.8	kPa	Assumed; replace with a cited value if found
Equipment point load (live), each main beam	P_eq	10	kN	Assumed
Guard load	-	cited only	-	Cite NBCC 2020 Div. B Part 4 guard clause / O. Reg. 213/91 guardrail section. NOT designed; flagged for review. Required per review R4; to be added at R1.
Interior joist tributary width	b_int	1.5	m	Formula: joist spacing
End joist tributary width	b_end	0.75	m	Formula: half joist spacing

LOAD COMBINATIONS / LIMITS	**Symbol**	**Value**	**Unit**	**Source / assumption**
Dead load factor (strength)	aD	1.25	-	1.25D + 1.5L
Live load factor (strength)	aL	1.5	-	1.25D + 1.5L
Deflection limit denominator (service, D+L)	L/	360	-	L/360 assumed

SCOPE LIMITS

1. Elastic Strength of Materials analysis only.
2. No CSA S16 resistance factors, no lateral-torsional buckling check, no column curve.
3. Connections, bracing and guard lateral load are not designed; flagged for senior review.
4. Loads are assumed unless a source is cited in column E.
5. Student exercise. Not for construction.

CHECK Three-way comparison

Quantity	Unit	Mathcad (hand)	Excel	Robot	% diff Excel vs hand	% diff Robot vs hand	PASS ($\leq$2%)	Cause if $>$2%
J1 max factored moment Mf	kN·m	13.51	13.515	13.51	0.0%	0.0%	PASS	Mathcad g = 9.807 m/s^2, Excel g = 9.81 m/s^2; under 0.05% effect on self-weight terms.
J1 max factored shear Vf	kN	18.02	18.020	18.02	0.0%	0.0%	PASS	
J1 service deflection	mm	3.561	3.561	3.56	0.0%	0.0%	PASS	
B1 reaction (factored)	kN	35.96	35.958	35.95	0.0%	0.0%	PASS	
B1 max factored moment Mf	kN·m	78.7	78.701	78.69	0.0%	0.0%	PASS	
B1 max factored shear Vf	kN	35.96	35.958	35.95	0.0%	0.0%	PASS	
B1 service deflection	mm	10.72	10.717	10.72	0.0%	0.0%	PASS	
C1 axial load (factored)	kN	46.03	46.028	46.83	0.0%	1.7%	PASS	Robot includes bracing self-weight (2.6 kN total; DL1 total 22.17 kN vs. 19.6 kN hand); +0.8 kN per post at ULS.
C1 Euler buckling load Pcr	kN	1546	1546.238	not run in robot	0.0%	N/A	N/A	
Total dead load	kN	19.6		22.17				bracing
Total live load	kN	106.4		106.4				

Review flags

- Long-side L64 braces sag about 24 mm under self-weight (L/280). Flag for senior review: consider a larger angle or a mid-length tie
- C1 member axial 41.9 kN (Robot) vs. 46.0 kN (hand): X-bracing shares about 4.9 kN of gravity load; the hand calc is conservative.

CALC Excel calculation (named inputs only)

J1 JOIST (interior)

Quantity	Value	Unit
wD dead UDL	0.971	kN/m
wL live UDL	7.200	kN/m
wf factored UDL	12.013	kN/m
Mf	13.515	kN·m
Vf	18.020	kN
sigma	85.001	MPa
tau	20.440	MPa
deflection (service)	3.561	mm
limit L/360	8.333	mm

B1 MAIN BEAM

Quantity	Value	Unit
PD per joist end	1.456	kN
PL per joist end	10.800	kN
Pf factored joist load	18.020	kN
Peq factored	15.000	kN
self-weight factored	0.476	kN/m
RB reaction	35.958	kN
MB midspan moment	78.701	kN·m
sigma	143.354	MPa
tau	19.999	MPa
Ps service joist load	12.256	kN
defl: joists at 1.5 and 4.5 m	4.456	mm
defl: midspan joist + equipment	5.884	mm
defl: self-weight	0.377	mm
deflection total	10.717	mm
limit L/360	16.667	mm

C1 POST (corner)

Quantity	Value	Unit
wD end joist	0.596	kN/m
wL end joist	3.600	kN/m
Re end joist reaction	9.217	kN
PC axial (factored)	46.028	kN
Pcr Euler	1546.238	kN
sigma axial	15.550	MPa
PC / Pcr	0.030	-

Values shown to 3 decimals; the live workbook (AS-P1-CAL-002.xlsx) holds full precision.

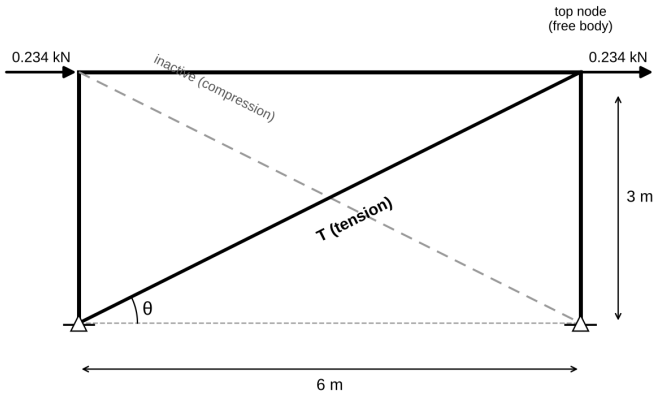
Purpose. Hand check of the R1 Robot model (AS-P1-MOD-001_R1) with BR1 braces (bars 18–25) set tension-only, under S16 notional loads and OBC guard loads, in response to review comments R1, R2 and R4. Robot reports axial force with compression positive; values below are tension unless noted.

Geometry and inputs. Long wall 6.0 m × 3.0 m: brace $L = \sqrt{6^2 + 3^2} = 6.708$ m, $\cos\theta = 0.894$, $\sin\theta = 0.447$, $\tan\theta = 0.5$. Short wall 3.0 m × 3.0 m: $L = 4.243$ m, $\cos\theta = \sin\theta = 0.707$, $\tan\theta = 1.0$. BR1 L64×64×6.4: A = 766 mm², 6.01 kg/m. C1: A = 2960 mm². E = 200 000 MPa. Gravity-only R1 post force (tension-only, ULS) = 46.55 kN.

1. S16 notional lateral load (Clause 8.4.1)

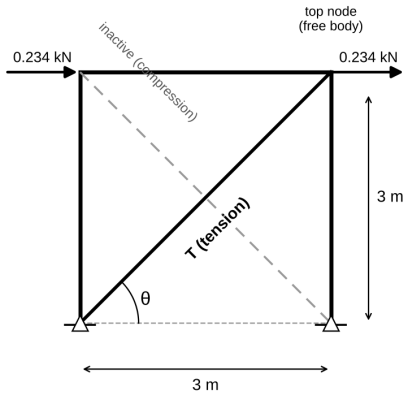
$N = 0.005 \times 187.32 = 0.94$ kN total (ULS gravity from Robot reactions), applied as 0.234 kN at each post top (nodes 5, 9, 10, 14). Load cases NX and NY, combined as 1.25D + 1.5L + 1.0N.

Long wall (6 m × 3 m), load in X



Brace length = √(6² + 3²) = 6.708 m
cos θ = 6 / 6.708 = 0.894
ΣFx at top node: T cos θ = 0.234 + 0.234 = 0.47 kN
T = 0.47 / 0.894 = 0.52 kN (Robot: 0.52)

Short wall (3 m × 3 m), load in Y



Brace length = √(3² + 3²) = 4.243 m
cos θ = 3 / 4.243 = 0.707
T cos θ = 0.47 kN
T = 0.47 / 0.707 = 0.66 kN (Robot: 0.66)

Each wall carries half the total (two parallel walls, symmetric frame): $H = 0.47$ kN per wall. At the loaded top node, $\Sigma F_x = 0$: $T\cos\theta = H$. Sway lengthens one diagonal of each X (tension, active); the other would shorten and is inactive.

	Long wall (NX)	Short wall (NY)
Brace tension $T = H/\cos\theta$	0.47/0.894 = **0.52 kN**	0.47/0.707 = **0.66 kN**
Brace self-weight $W = L \times 6.01 \times 9.81/1000$	0.395 kN	0.250 kN
Self-weight component along brace, ULS: $1.25\,W\sin\theta$	0.22 kN	0.22 kN
Tension at top end	0.52 + 0.22 = **0.74 kN**	0.66 + 0.22 = **0.88 kN**
Overturning $M = 0.94 \times 3.0 = 2.82$ kN·m; per post $M/b/2$	2.82/6/2 = 0.24 kN	2.82/3/2 = 0.47 kN
Post force $46.55 + \Delta$	**46.79 kN**	**47.02 kN**

2. Guard lateral load (OBC 4.1.5.14)

Guard load 0.75 kN/m outward at guard height (OBC 2024 Div. B, 4.1.5.14, “other locations”; category to be confirmed by reviewer). Applied as a horizontal line load on one edge only, which is the worst case for the bracing: GX on the 6 m-end joist (3 m long, bar 17); GY on one B1 line (6 m, bars 9–12). Combinations ULS+GX = 1.25D + 1.5L + 1.5GX + 1.0NX and ULS+GY = 1.25D + 1.5L + 1.5GY + 1.0NY. Combining full floor live load with guard load is conservative and is noted for review. The loaded edge member delivers its load to two posts, one on each parallel wall, so each wall carries half.

	ULS+GX (long walls)	**ULS+GY (short walls)**
Factored guard load $1.5 \times 0.75\times$ edge length	$1.125 \times 3 = 3.375$ kN	$1.125 \times 6 = 6.75$ kN
Total lateral (guard + notional 0.94)	**4.31 kN**	**7.69 kN**
Per wall H	2.16 kN	3.85 kN
Brace tension $T = H/\cos\theta$	2.16/0.894 = **2.41 kN**	3.85/0.707 = **5.44 kN**
Tension at top end (+ 0.22)	**2.63 kN**	**5.66 kN**
Overturning $M = H_{tot} \times 3.0$	12.94 kN·m	23.07 kN·m
Per post $M/b/2$	12.94/6/2 = 1.08 kN	23.07/3/2 = 3.85 kN
Post force $46.55 + \Delta$	**47.63 kN**	**50.40 kN**

3. SLS displacements (comment R1)

SLS combinations: SLS+GX = D + L + GX and SLS+GY = D + L + GY (unfactored, notional loads omitted).

- **Post shortening (gravity):** $\delta_p = PL/AE = 32.14 \times 10^3 \times 3000/(2960 \times 200\,000) = 0.16$ mm, using the SLS reaction 32.14 kN.
- **Brace slack.** Gravity lowers the post tops by δ_p, which shortens each diagonal by $\delta_p \sin\theta$. The tension brace only engages after the frame has swayed far enough to recover this: horizontal slack $= \delta_p \tan\theta$.
- **Brace stretch.** $\delta_b = TL/AE$; horizontal component $\delta_b/\cos\theta$.

	SLS+GX (long wall)	**SLS+GY (short wall)**
Service guard load per wall	$0.75 \times 3/2 = 1.125$ kN	$0.75 \times 6/2 = 2.25$ kN
Brace tension T	1.26 kN	3.18 kN
Brace stretch $\delta_b = TL/AE$	0.055 mm	0.088 mm
Horizontal: $\delta_b/\cos\theta$	0.062 mm	0.125 mm
Slack: $\delta_p \tan\theta$	0.16 × 0.5 = 0.081 mm	0.16 × 1.0 = 0.163 mm
Frame drift at post top	**0.14 mm**	**0.29 mm**

B1 lateral deflection (SLS+GY). The joists tie both B1 lines together, so they bend about their weak axes as a pair: $\delta = 5wL^4/(384E \cdot 2I_y) = 5 \times 0.75 \times 6000^4/(384 \times 200\,000 \times 2 \times 7.23 \times 10^6) = 4.38$ mm, plus 0.29 mm frame drift at the supports = **4.67 mm**. (I_y = 7.23×10^6 mm^4 for W310×39, CISC.)

B1 vertical deflection (SLS+GY). 10.72 mm (AS-P1-CAL-001) + 0.16 mm post shortening = **10.88 mm**.

4. Hand vs Robot

#	Quantity	Case	Hand	Robot	Diff
1	Brace T, long wall (bars 18, 20), bottom / top (kN)	ULS+NX	0.52 / 0.74	0.52 / 0.74	0.0%
2	Brace T, short wall (bars 22, 24), bottom / top (kN)	ULS+NY	0.66 / 0.88	0.66 / 0.88	0.0%
3	Post force (kN)	ULS+NX	46.79	46.79	0.0%
4	Post force (kN)	ULS+NY	47.02	47.02	0.0%
5	Brace T, long wall, bottom / top (kN)	ULS+GX	2.41 / 2.63	2.41 / 2.63	0.0%
6	Brace T, short wall, bottom / top (kN)	ULS+GY	5.44 / 5.66	5.43 / 5.66	0.2%
7	Post force (kN)	ULS+GX	47.63	47.63	0.0%
8	Post force (kN)	ULS+GY	50.40	50.40	0.0%
9	Post shortening (mm)	SLS	0.16	0.16	0.0%
10	Frame drift at post top (mm)	SLS+GX	0.14	0.15	−7%
11	Frame drift at post top (mm)	SLS+GY	0.29	0.30	−3%
12	B1 lateral at midspan (mm)	SLS+GY	4.67	4.65	+0.4%
13	B1 vertical at midspan (mm)	SLS+GY	10.88	10.88	0.0%

Drift differences of a few hundredths of a millimetre come from rounding and from minor flexibility not included by hand (beam axial shortening, connection eccentricities); they are not significant.

5. Results carried forward

- Under gravity alone all braces are inactive and the posts carry the full load (46.55 kN). The R0 finding of about 4.9 kN through the bracing was a result of the elastic model and is closed (R2).
- Under lateral load exactly one brace per X is active, in tension, and every force above is confirmed by hand (R1, R4).
- **Governing brace force for AS-P1-CAL-004: T_f = 5.66 kN** (bars 22/24, ULS+GY).
- **Governing post force for AS-P1-CAL-004: C_f = 50.40 kN** (ULS+GY).
- SLS frame drift 0.30 mm (H/10 000); B1 lateral 4.65 mm (L/1290). Both are far below any applicable limit.

AS-P1-CAL-004 | Rev R1 | A. Saleh | 2026-10-21
CSA S16-19 member checks: B1, J1, C1, BR1.
Student exercise, not for construction.

$F_y := 350 \cdot MPa$
$E := 200000 \cdot MPa$ 350W assumed (CSA G40.21). S16 Cl. 13.1
$\phi := 0.9$
$F_u := 450 \cdot MPa$
$G := 77000 \cdot MPa$
$\phi_u := 0.75$

B1, W310×39 (Cl. 11, 13.4.1.1, 13.6, 13.8)

$d_B1 := 310 \cdot mm$ $b_B1 := 165 \cdot mm$ $t_B1 := 9.7 \cdot mm$
$w_B1 := 5.8 \cdot mm$
$Z_x.B1 := 610 \cdot 10^3 \cdot mm^3$
$I_y.B1 := 7.27 \cdot 10^6 \cdot mm^4$
$J_B1 := 126 \cdot 10^3 \cdot mm^4$
$C_w.B1 := 164 \cdot 10^9 \cdot mm^6$

$A_B1 := 4940 \cdot mm^2$
$I_x.B1 := 85.1 \cdot 10^6 \cdot mm^4$

$$Fy_n := \frac{F_y}{MPa}$$

$$flangeB1 := \frac{b_B1}{(2 \cdot t_B1)}$$

$flangeB1 = 8.505$

$$limit1 := \frac{145}{\sqrt[2]{Fy_n}}$$

$limit1 = 7.751$

$$limit2 := \frac{170}{\sqrt[2]{Fy_n}}$$

$limit2 = 9.087$

$hB1 := d_B1 - 2 \cdot t_B1$

$$webB1 := \frac{hB1}{w_B1}$$

$webB1 = 50.103$

$$limitw1 := \frac{1100}{\sqrt[2]{Fy_n}}$$

$$limitw1 = 58.797$$

Flange Class 2, web Class 1, so section Class 2 (governed by the worse of the two). Use Z_x for Mr

$$limitV := \frac{1014}{\sqrt[2]{Fy_n}} \qquad limitV = 54.201$$

$$VrB1 := \phi \cdot d_B1 \cdot w_B1 \cdot 0.66 \cdot F_y \qquad VrB1 = 373.804\ \boldsymbol{kN}$$

$$MfB1 := 78.70 \cdot \boldsymbol{kN} \cdot \boldsymbol{m}$$
$$VfB1 := 35.96 \cdot \boldsymbol{kN}$$

$$ratioVB1 := \frac{VfB1}{VrB1} \qquad ratioVB1 = 0.096$$

$$MpB1 := Z_x.B1 \cdot F_y \qquad MpB1 = 213.5\ \boldsymbol{kN} \cdot \boldsymbol{m}$$

$$omega2 := 1 \qquad omega2 = 1$$

$$LuB1 := 1.5 \cdot \boldsymbol{m} \qquad LuB1 = 1.5\ \boldsymbol{m}$$

$$term1 := E \cdot I_y.B1 \cdot G \cdot J_B1$$

$$k := \pi \cdot \frac{E}{LuB1}$$

$$term2 := k \cdot k \cdot I_y.B1 \cdot C_w.B1$$

$$MuB1 := omega2 \cdot \left(\frac{\pi}{LuB1}\right) \cdot \sqrt[2]{term1 + term2}$$

$$MuB1 = 989.707\ \boldsymbol{kN} \cdot \boldsymbol{m}$$

$$MrB1 := \text{if}\left(MuB1 > 0.67 \cdot MpB1,\ min\left(1.15 \cdot \phi \cdot MpB1 \cdot \left(1 - 0.28 \cdot \frac{MpB1}{MuB1}\right),\ \phi \cdot MpB1\right),\ \phi \cdot MuB1\right)$$

$MrB1 = 192.15\ \mathbf{kN \cdot m}$

$$ratioMB1 := \frac{MfB1}{MrB1}$$

$ratioMB1 = 0.41$

$LuU := 6 \cdot \mathbf{m}$

$$kU := \pi \cdot \frac{E}{LuU}$$

$term2U := kU \cdot kU \cdot I_y.B1 \cdot C_w.B1$

$$MuB1u := omega2 \cdot \left(\frac{\pi}{LuU}\right) \cdot \sqrt[2]{term1 + term2U}$$

$MuB1u = 86.325\ \mathbf{kN \cdot m}$

$$MrB1u := \text{if}\left(MuB1u > 0.67 \cdot MpB1,\ min\left(1.15 \cdot \phi \cdot MpB1 \cdot \left(1 - 0.28 \cdot \frac{MpB1}{MuB1u}\right),\ \phi \cdot MpB1\right),\ \phi \cdot MuB1u\right)$$

$MrB1u = 77.692\ \mathbf{kN \cdot m}$

$$ratioMB1u := \frac{MfB1}{MrB1u}$$

$ratioMB1u = 1.013$

If J1 does not restrain the B1 compression flange, Lu = 6.0 m and B1 fails (1.01). Lateral restraint at the joists is required (Finding 8).

$Z_y.B1 := 135 \cdot 10^3 \cdot \mathbf{mm}^3$

$MryB1 := \phi \cdot Z_y.B1 \cdot F_y \qquad MryB1 = 42.525\ \mathbf{kN \cdot m}$

$$wG := 1.125 \cdot \frac{\mathbf{kN}}{\mathbf{m}}$$

$L6 := 6 \cdot \mathbf{m}$

$MfyB1 := wG \cdot L6 \cdot \frac{L6}{8}$ $MfyB1 = 5.063\ kN \cdot m$

$rx := \frac{MfB1}{MrB1}$

$ry := \frac{MfyB1}{MryB1}$

$ratioBiax := rx + ry$

$ratioBiax = 0.529$

J1, W150×22 (Cl. 11, 13.5(c), 13.4.1.1, 13.6

$d_J1 := 152 \cdot mm$ $b_J1 := 152 \cdot mm$ $t_J1 := 6.6 \cdot mm$ $w_J1 := 5.8 \cdot mm$

$I_x.J1 := 12.1 \cdot 10^6 \cdot mm^4$ $I_y.J1 := 3.87 \cdot 10^6 \cdot mm^4$

$J_J1 := 43.4 \cdot 10^3 \cdot mm^4$ $C_w.J1 := 20.4 \cdot 10^9 \cdot mm^6$

$MfJ1 := 13.51 \cdot kN \cdot m$ $VfJ1 := 18.02 \cdot kN$

$flangeJ1 := \frac{b_J1}{(2 \cdot t_J1)}$ $flangeJ1 = 11.515$

$limit3 := \frac{200}{\sqrt[2]{Fy_n}}$ $limit3 = 10.69$

$hJ1 := d_J1 - 2 \cdot t_J1$

$webJ1 := \frac{hJ1}{w_J1}$ $webJ1 = 23.931$

Flange Class 4 → effective flange width per S16 Cl. 13.5(c).

$be := 400 \cdot \frac{t_J1}{\sqrt[2]{Fy_n}}$ $be = 141.114\ mm$

$dA := (b_J1 - be) \cdot t_J1$

$$yf := \frac{(d_J1 - t_J1)}{2}$$

$$Ie := I_xJ1 - 2 \cdot dA \cdot yf \cdot yf$$

$$Se := \frac{Ie}{\left(\frac{d_J1}{2}\right)} \qquad Se = \left(1.492 \cdot 10^5\right)\ mm^3$$

$$MyJ1 := Se \cdot F_y \qquad MyJ1 = 52.226\ kN \cdot m$$

$$VrJ1 := \phi \cdot d_J1 \cdot w_J1 \cdot 0.66 \cdot F_y \qquad VrJ1 = 183.285\ kN$$

$$ratioVJ1 := \frac{VfJ1}{VrJ1} \qquad ratioVJ1 = 0.098$$

$$LuJ1 := 3 \cdot m$$

$$term1J := E \cdot I_yJ1 \cdot G \cdot J_J1$$

$$kJ := \pi \cdot \frac{E}{LuJ1}$$

$$term2J := kJ \cdot kJ \cdot I_yJ1 \cdot C_wJ1$$

$$MuJ1 := omega2 \cdot \left(\frac{\pi}{(LuJ1)}\right) \cdot \sqrt[2]{term1J + term2J} \qquad MuJ1 = 81.45\ kN \cdot m$$

$$MrJ1 := \text{if}\left(MuJ1 > 0.67 \cdot MyJ1,\ min\left(1.15 \cdot \phi \cdot MyJ1 \cdot \left(1 - 0.28 \cdot \frac{MyJ1}{MuJ1}\right),\ \phi \cdot MyJ1\right),\ \phi \cdot MuJ1\right)$$

$$MrJ1 = 44.349\ kN \cdot m$$

$$ratioMJ1 := \frac{MfJ1}{MrJ1} \qquad ratioMJ1 = 0.305$$

Class 4: My = Se·Fy used in place of Mp. J1 taken as unbraced over 3.0 m (no restraint from grating).

C1, HSS127×127×6.4 (Cl.13.3.1):

$A_C1 := 2960 \cdot mm^2$ $r_C1 := 48.8 \cdot mm$ $b_C1 := 127 \cdot mm$ $t_C1 := 6.35 \cdot mm$

$CfC1 := 50.40 \cdot kN$ $L_C1 := 3 \cdot m$ $K := 1$

$belt := \frac{(b_C1 - 4 \cdot t_C1)}{t_C1}$ $belt = 16$

$KLr := K \cdot \frac{L_C1}{r_C1}$ $KLr = 61.475$

$ratioFe := \frac{Fy_n}{(\pi \cdot \pi \cdot 200000)}$

$Slend := KLr \cdot \sqrt[2]{ratioFe}$

$Slend = 0.819$

$n := 1.34$

$p1 := \exp(n \cdot 2 \cdot \ln(Slend))$

$base := 1 + p1$

$fac := \exp\left(\frac{-\ln(base)}{n}\right)$

$CrC1 := \phi \cdot A_C1 \cdot F_y \cdot fac$

$CrC1 = 661.244\ kN$

$ratioC1 := \frac{CfC1}{CrC1}$

$ratioC1 = 0.076$

Cr = φAFy(1 + λ^2n)^(-1/n), n = 1.34 (Class C HSS), S16 Cl. 13.3.1. Written with exp/ln in Mathcad.

BR1, L64×64×6.4 and M20 A325 bolt (Cl. 13.2, 12.3.3, 13.12.1.2)

$A_BR := 768 \cdot mm^2$

$t_BR := 6.35 \cdot mm$

$TfBR := 5.66 \cdot kN$

$F_ub := 825 \cdot MPa$

$A_b := 314 \cdot mm^2$

$d_b := 20 \cdot mm$

$TrG := \phi \cdot A_BR \cdot F_y$

$TrG = 241.92\ kN$

$An := A_BR - 24 \cdot mm \cdot t_BR$
$Ane := 0.60 \cdot An$

$TrN := \phi_u \cdot Ane \cdot F_u$

$TrN = 124.659\ kN$

$VrBolt := 0.6 \cdot 0.8 \cdot A_b \cdot F_ub \cdot 0.7$

$VrBolt = 87.041\ kN$

$BrBolt := 3 \cdot 0.8 \cdot t_BR \cdot d_b \cdot F_u$

$BrBolt = 137.16\ kN$

$ratioBR := \frac{TfBR}{VrBolt}$

$ratioBR = 0.065$

Assumed 1 M20 A325 bolt per brace end, threads intercepted. Shear lag 0.60An (angle connected by one leg, fewer than 4 bolts). Bolt shear governs.

SUMMARY: CSA S16 MEMBER CHECKS	Factored	Resistance	Ratio	
B1 Bending, Lu = 1.5 m	78.70 kN·m	192.2 kN·m	0.41	
B1 Shear	35.96 kN	373.8 kN	0.10	
B1 Biaxial bending (ULS+GY)			0.53	
B1 Bending, Lu = 6.0 m (unbraced)	78.70 kN·m	77.7 kN·m	1.01	FAILS: needs J1 restraint
J1 Bending, Class 4, Lu = 3.0 m	13.51 kN·m	44.3 kN·m	0.30	
J1 Shear	18.02 kN	183.3 kN	0.10	
C1 Compression	50.40 kN	661.2 kN	0.08	
BR1 Tension, gross	5.66 kN	241.9 kN	0.02	
BR1 Tension, net	5.66 kN	124.7 kN	0.05	
BR1 Bolt shear (governs)	5.66 kN	87.1 kN	0.07	
BR1 Bolt bearing	5.66 kN	137.2 kN	0.04	

All members pass with J1 providing lateral restraint to B1.

AS-P1-MTO-001 Material Take-Off, Maintenance Platform (Portfolio Study)

Rev R1, 2026-10-21. Lengths from Revit schedules (AS-P1-MOD-002, sheet S-301). Mass/m from CISC Handbook. Grey cells = inputs.

Mark	Member	Section (CISC)	Revit type	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
B1	Main beam	W310x39	W12X26	2	5.848	11.696	38.80	453.8	Revit cut length
J1 (end)	Joist	W150x22	W6X15	2	2.848	5.696	22.50	128.2	J1-1, J1-5
J1 (int.)	Joist	W150x22	W6X15	3	2.810	8.430	22.50	189.7	J1-2 to J1-4
C1	Post	HSS127x127x6.4	HSS5X5X1/4	4	3.000	12.000	23.20	278.4	System length GRADE to T/S
BR1 (long)	X-brace, long sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	6.463	25.852	6.01	155.4	BR1-1 to BR1-4
BR1 (short)	X-brace, short sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	3.848	15.392	6.01	92.5	BR1-5 to BR1-8
Subtotal, members				**19**		**79.066**		**1297.9**	
Connection allowance	Plates, bolts, welds						10%	129.8	ASSUMED % of member mass, not from code. Confirm with reviewer.
TOTAL STEEL								**1427.7**	**kg**

Cross-check against Revit schedule volumes

Item	Value	Unit	Source
Framing volume	0.1336	m3	P1 Steel Framing Schedule, grand total
Column volume	0.0355	m3	P1 Column Schedule, grand total
Steel density	7850.0	kg/m3	Standard value for structural steel
Mass from Revit volume	1327.4	kg	Formula
Mass from CISC kg/m	1297.9	kg	Members subtotal above
Difference	2.3%		Expected small: Revit families use imperial section geometry
Status	WITHIN 5% - OK		Tolerance 5% (own criterion)

BR1 Slenderness Check, L64×64×6.4 X-Bracing

Lengths are work-point (centre-to-centre) geometry, conservative versus Revit cut length. Grey cells = inputs.

r_min (mm)	12.5	Computed: $\sqrt{}$(Iyp/A), Iyp and A from beamdimensions.com L64x64x6.4
Tension limit L/r	300	CSA S16 slenderness limit, tension members
Compression limit KL/r	200	CSA S16 slenderness limit, compression members

Brace	Panel w (mm)	Panel h (mm)	L full (mm)	L/r full	L at crossing (mm)	L/r at crossing	Tension-only check	If designed for compression
Long sides (BR1-1 to 4)	6000	3000	6708	537	3354	268	OK if bolted at crossing	FAIL - keep tension-only
Short sides (BR1-5 to 8)	3000	3000	4243	339	2121	170	OK if bolted at crossing	OK

Finding: Full-length L/r exceeds the tension limit on both brace types. Braces must be bolted at the X intersection (halves unbraced length) and treated as tension-only. Add note to S-101 and brief. (Done at R1: tension-only bracing, bolted crossing, half-length L/r 268 and 170, both $\leq$ 300.)

AS-P1-REG-001 Review and Issue Register

Rev R1, 2026-10-21

No.	Date	Source	Item	Finding	Action	Status
1	2026-10-18	Revit schedule	BR1 section	Braces modelled as W6X15. Schedule volume check gave ∼2950 mm2 area vs ∼768 mm2 expected for L64x64x6.4.	Type changed to L2-1/2X2-1/2X1/4 (L64x64x6.4 equivalent). Schedule re-checked.	Closed
2	2026-10-18	Revit schedule	J1 structural usage	Joists tagged as Girder.	Changed to Joist.	Closed
3	2026-10-18	Revit model	BR1 placement method	Braces drawn with Beam tool, so Structural Usage cannot be set to Vertical Bracing.	Accepted for this model. Use Brace tool on future models.	Closed
4	2026-10-18	Drawing S-101 R0	C1 label	Section labelled HSS128x128.	Corrected to HSS127x127x6.4 in R1.	Closed
5	2026-10-18	Drawing S-101 R0	BR1 label	Typo 'BR!'.	Corrected to BR1 in R1.	Closed
6	2026-10-18	Drawing S-101 R0	T/S label	Top-of-steel label poorly placed.	Moved in R1.	Closed
7	2026-10-18	Robot vs hand calc	C1 reaction	Robot 46.83 kN vs hand 46.03 kN.	Difference = bracing self-weight, included in Robot only. Accepted.	Closed
8	2026-10-18	Robot vs hand calc	C1 axial force	Robot member force 41.9 kN, lower than hand calc.	R1 tension-only run: braces inactive under gravity; post force 46.55 kN. Gap was elastic brace participation.	Closed
9	2026-10-18	Slenderness check	BR1 L/r	Full-length L/r exceeds tension limit (see SLENDERNESS tab).	Tension-only adopted; half-length L/r 268 $\leq$ 300. Noted on S-101/S-201 Rev 2–3.	Closed
10	2026-10-18	Robot	BR1 self-weight sag	Brace sag ∼24 mm (about L/280) under self-weight.	Intersection connection reduces it. Mention in brief.	Open
11	2026-10-18	MTO	Mass cross-check	Revit volume mass vs CISC kg/m mass differ by a few %.	Imperial family geometry. Within 5% tolerance. Accepted.	Closed
12	2026-10-18	Revit type	C1 section area	Earlier note said Revit HSS area 2774 mm2 vs CISC 2960. Column volume implies ∼2967 mm2.	Checked: column schedule volume 0.0089 m3 / 3.0 m = 2967 mm2, matches CISC 2960 mm2.	Closed
13	2026-10-21	CISC check	BR1 area	R0 used 766 mm2; CISC gives 768 mm2 (2L64x64x6.4 = 1536 mm2).	R1 uses 768 mm2. Difference 0.3%, no effect on R0 results.	Closed
14	2026-10-21	S16 check (CAL-004)	J1 flange class	b/2t = 11.5 exceeds Class 3 limit 200/sqrt(Fy) = 10.7; J1 flange is Class 4 in 350W.	Effective flange width 400t/sqrt(Fy) = 141 mm used (S16 Cl. 13.5(c)); Mr = 44.3 kN.m, utilization 0.30.	Closed
15	2026-10-21	S16 check (CAL-004)	B1 lateral support	B1 Mr assumes compression flange braced by J1 at 1.5 m (utilization 0.41). Unbraced over 6.0 m, utilization 1.01.	Joist-to-beam connection to provide lateral restraint; confirm in connection design.	Open

External review comments (record only comments actually received)

No.	Date	Reviewer	Comment (verbatim)	Response	Action	Status
R1	2026-10-18	S. Al-Zubaidi, P.Eng.	“I’d suggest modelling the braces as tension-only members, applying the relevant lateral loads, and checking the resulting member forces and displacements.”	Braces set tension-only (bars 18–25). Notional (S16 8.4.1) and guard (NBC 2020 4.1.5.14) lateral cases run; forces hand-checked. SLS drift 0.30 mm (H/10000); B1 lateral 4.65 mm (L/1290), hand-checked. See AS-P1-CAL-003, AS-P1-MOD-001_R1.	Set BR1 tension-only in Robot; add lateral load case	Closed
R2	2026-10-18	S. Al-Zubaidi, P.Eng.	“What I would check first is whether Robot is allowing any of the braces to carry compression when you’ve intended them to act in tension only.”	Confirmed: under gravity Robot sets all braces inactive; post force 46.55 kN vs reaction 46.83 kN. See AS-P1-CAL-003.	Re-run with tension-only braces; explain the 41.9 vs 46.03 kN gap. Keep 46.03 kN for now.	Closed
R3	2026-10-18	S. Al-Zubaidi, P.Eng.	“For your next revision, I’d suggest identifying the relevant code provisions and explaining why you’ve selected those criteria.”	Cited in brief R1: live load 3.6 kPa minimum for equipment areas (NBC 2020 Div. B Table 4.1.5.3, adopted by O. Reg. 163/24), 4.8 kPa adopted for laydown; deflection L/360 adopted vs L/300 recommended (CSA S16 Annex D).	Cite the OBC live-load and deflection provisions in the design basis	Closed
R4	2026-10-18	S. Al-Zubaidi, P.Eng.	“I’d include the guard loading in your lateral analysis and then check the resulting member forces and connections.”	Guard 0.75 kN/m outward on one edge (NBC 2020 4.1.5.14). Governing brace Tf = 5.66 kN, post Cf = 50.40 kN (ULS+GY). Members and brace end bolt checked in AS-P1-CAL-004 (bolt shear governs, 0.07). Other connections not designed.	Add guard load case (OBC); check brace forces and drift	Closed
R5	2026-10-18	S. Al-Zubaidi, P.Eng.	“I don’t have an issue with using 10% as a preliminary allowance for connections at this stage.”		Keep it labelled as an estimate	Closed
R6	2026-10-18	S. Al-Zubaidi, P.Eng.	“They don’t replace the resistance checks required by CSA S16.”	S16 checks in AS-P1-CAL-004: class, Vr, Mr (LTB), Cr, Tr, bolt. All members pass; max utilization 0.53 (B1 biaxial). B1 relies on J1 restraint (row 15).	Add his S16 checklist to the brief as “Not yet checked”	Closed